$P_B(\cdot) = P(\cdot|B) \Rightarrow P_B(A|C) = P(A|B \cap C)$

$$P_B(A|C) = \frac{P_B(A \cap C)}{P_B(C)} = \frac{P(A \cap C|B)}{P(C \cap B)/P(B)} = \frac{P(A \cap B \cap C)}{\cancel{P(B)} \cdot P(C \cap B)/\cancel{P(B)}}$$

$$= P(A|B \cap C)$$

$P(A|B) = \alpha P(A|B \cap C) + (1-\alpha) P(A|B \cap C^c)$; $C \subset B$; $\alpha = ?$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A|B \cap C) P(B \cap C) + P(A|B \cap C^c) P(B \cap C^c)}{P(B)}$$

$P(B \cap C^c) + P(B \cap C) = P(B)$

$$= P(A|B \cap C) \frac{P(C)}{P(B)} + P(A|B \cap C^c) \frac{P(B) - P(C)}{P(B)}$$

$$\Rightarrow \alpha = \frac{P(C)}{P(B)}$$

$P(A \cup B) = 0.4 \quad P(A \cap B) = 0.1 \quad P(A) + P(B) = 0.5$

$0.1 < P(A), P(B) < 0.4$

$P(\text{wet}|\text{rained}) = 0.8 \quad P(\text{wet}|\text{tank}) = 0.5$

$P(\text{wet}|\text{neither}) = 0.1 \quad P(\text{rain}) = 0.2 \quad P(\text{truck}) = 0.4$

$P(\text{neither}) = 0.5$

$P(\text{rain} \cup \text{truck}) = 1 - 0.5 = 0.5 \quad P(T \cap R') = 0.4 - 0.1 = 0.3$

$P(\text{rain} \cap \text{truck}) = 0.1$

$$P(R|W) = \frac{P(W|R) P(R)}{P(W)} = \frac{0.8 \times 0.2}{0.8 \times 0.2 + 0.5 \times 0.3 + 0.5 \times 0.1} = \frac{4}{9}$$

10 numbered balls, 2 unnumbered, draw them one-by-one until both unnumbered appears. Size of sample space = ?

sps we stop at k^{th} pick $(k \geq 2, \leq 12)$

$(k-1)\ {}^{10}P_{(k-2)} = \frac{10!}{(12-k)!} \times (k-1)$

$(k-1)$ ↓ for first unnumbered ball

$$S.S \Rightarrow \sum_{k=2}^{12} (k-1)\frac{10!}{(12-k)!} = \sum_{k=2}^{12} {}^{10}C_{k-2}\ (k-1)!$$

$X \rightarrow 3R+4B \qquad Y \rightarrow 4R+5B$

(i) $\frac{1}{2} \times \frac{3}{7} \times \left[\frac{1}{2} \times \frac{3}{8} + \frac{1}{2} \times \frac{4}{9}\right] + \frac{1}{2} \times \frac{4}{7} \times [\ldots]$